

CLIENT BRIEFING DOCUMENT

Impact Intelligence Framework (IIF)

What we built, how it works, and how it delivers on the MoU

PREPARED BY

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DATE

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PREPARED FOR

Uptonville Nigeria Limited

VERSION

Phase 1 — Complete Build

This document summarises the Phase 1 deliverables of the Impact Intelligence Framework, built under the Strategic Collaboration Agreement between Impact Driver and Uptonville Nigeria Limited.

01 Executive Summary

The Impact Intelligence Framework (IIF) is a co-branded, full-stack web platform that digitises, structures, and analyses ESG and environmental impact data from Uptonville Nigeria Limited's historical project portfolio. It transforms unstructured project records into a live, queryable knowledge base that powers eight analytical engines — producing regulatory disclosures, benchmarking insights, and decision support intelligence for future projects.

What the partnership set out to achieve

The Strategic Collaboration Agreement between Impact Driver and Uptonville Nigeria Limited established a mandate to build a digitised ESG and impact reporting platform tailored to the Nigerian oil and gas sector. The platform was to ingest data from 7–10 historical Uptonville projects, structure it against a standardised 12-section extraction template, and aggregate it into eight analytical engines covering regulatory compliance, stakeholder intelligence, decision support, benchmarking, and disclosure reporting.

What Phase 1 delivers

- A complete full-stack web application — accessible from any device, secured behind role-based authentication
- A 12-section project template (Sections A–L) that captures every ESG data point the MoU specifies
- Eight live analytical engines that aggregate data across all projects as new templates are filled
- A disclosure reporting engine that generates co-branded PDF reports aligned to GRI, ISSB, IFC, TNFD, and NUPRC frameworks
- An admin panel for user management, CSV data exports, and system statistics
- Full deployment infrastructure for Vercel (frontend) and Render (backend) with MongoDB Atlas as the database

02 Platform Architecture

The platform is built as a modern decoupled web application: a React/Next.js frontend served from Vercel communicates with a Node.js/Express API on Render, which reads and writes to a MongoDB Atlas database. The frontend and backend are maintained in separate GitHub repositories and deployed independently.

LAYER	TECHNOLOGY	DEPLOYMENT	PURPOSE
Frontend	Next.js 14, TypeScript, Tailwind CSS	Vercel (iad1 region)	All user-facing screens — project register, extraction workspace, engine dashboards, admin panel
Backend API	Node.js, Express, TypeScript	Render (web service)	REST API serving all data operations, PDF generation, CSV export, auth
Database	MongoDB Atlas	Cloud (dedicated cluster)	Stores all project records, templates, users, engine maps, sign-offs
PDF Engine	pdfkit (pure Node.js)	Runs on Render	Server-side PDF generation — no Chromium dependency, no headless browser
Auth	JWT (7-day tokens)	Stateless, backend-enforced	Role-based access control with section-level permission gates

Security and access control

Every API route is protected by JWT authentication middleware. Section-level write permissions are enforced on both the frontend (UI gating) and backend (controller-level role checks), ensuring that, for example, an Impact Driver Analyst cannot overwrite sections owned by the Uptonville Technical Reviewer.

ROLE	SECTIONS (WRITE)	SECTIONS (READ)	ADDITIONAL ACCESS
uptonville_reviewer	A, D, G, H	All	Project creation, project register
impact_driver_analyst	B, C, E, F, I, J	All	Engine dashboards, report generation
steering_committee	L (sign-off only)	All	Can set status to signed-off
admin	All	All	User management, CSV exports, system stats, backfill utilities

03 The 12-Section Project Template

Every project on the platform is linked to a single structured extraction template containing twelve sections (A–L). Sections A–J correspond directly to the data capture framework defined in the MoU. Sections K and L handle engine mapping and formal sign-off. Each section tracks its own completion status (not started / in progress / complete) and is protected by role-based access.

SECTION	TITLE	ROLE (WRITE)	FEEDS ENGINE(S)	KEY DATA CAPTURED
A	Project Identification & Context	Uptonville Reviewer	Project Classification, Benchmarking	Name, reference code, type, location, environment, client, operator, duration, value tier
B	Indicator Library	Analyst	Indicator Library, Materiality, Benchmarking	ESG indicators (E/S/G), units, measurement methods, rationale
C	Regulatory Rules	Analyst	Regulatory Rules Engine	Regulations applied, issuing body, application details (NUPRC, NESREA, NOSDRA, IFC, etc.)
D	Stakeholder Registry	Uptonville Reviewer	Stakeholder Intelligence, Materiality	Stakeholder groups, interests, reporting format required, engagement outcomes
E	Mitigation Measures (Decision Support)	Analyst	Decision Support Engine	Mitigation measures, effectiveness (high/medium/low), evidence, recommendations, expert reasoning
F	Regulatory Evidence	Analyst	Regulatory Rules Engine, Reporting	Evidence submitted per regulation, format, frequency, accepted without dispute (Y/N)
G	Environmental Outcomes	Uptonville Reviewer	Materiality, Benchmarking, Reporting	Outcomes achieved, measurement method, timeframe, outstanding issues
H	Social Impacts	Uptonville Reviewer	Materiality, Stakeholder Intelligence	Positive/negative impacts, grievance mechanism, grievance outcomes
I	Disclosure Alignment	Analyst	Reporting Engine	Disclosure topics mapped to GRI/ISSB/IFC/TNFD/NUPRC frameworks with value rationale
J	Automation Opportunities	Analyst	Platform Backlog	Data collection pain points, manual processes, automation opportunities, priority ratings
K	Engine Contribution Map	Both roles	All engines	Tick-box confirmation of which engines this project contributed to; most valuable insight per engine
L	Validation & Sign-off	Steering Committee	Governance record	Three-signature sign-off: Uptonville Technical Reviewer, Impact Driver Analyst, Joint Steering Committee

04 The Eight Platform Engines

The eight engines are the analytical core of the platform. Each engine is a MongoDB aggregation pipeline that runs across all completed project templates in real time — there is no separate data entry step. As analysts fill in templates, the engines automatically incorporate the new data. Engines are accessible from the main navigation and display live results with visualisations.

ENGINE	DATA SOURCE	WHAT IT PRODUCES
01 — Project Classification Engine	Section A across all projects	Taxonomy of project types, operating environments, and statuses. Counts by category. Feeds the project register overview cards.
02 — Regulatory Rules Engine	Sections C and F	A cross-project library of regulations, issuing bodies, and how each has been applied. Identifies which regulations are most common. Shows evidence acceptance rates.
03 — Indicator Library	Section B	Aggregated ESG indicator catalogue with unit standardisation, measurement method comparison, and near-duplicate detection (Dice coefficient string similarity). Filterable by category and project type.
04 — Materiality Engine	Sections B, D, G, H	Materiality matrix: maps project types and stakeholder groups to the ESG topics that matter most, informed by real outcomes rather than generic frameworks.
05 — Stakeholder Intelligence Engine	Section D	Profiles of recurring stakeholder groups across projects — their concerns, reporting formats they need, and engagement outcomes. Supports planning for future projects.
06 — Decision Support Engine	Section E	Cross-project library of mitigation measures with effectiveness ratings and expert reasoning. Filterable by effectiveness and keyword search. The most-queried engine.
07 — Benchmarking Engine	Sections A, B, G, H	ESG coverage depth by project type (stacked bar chart) and unique indicator counts per type (horizontal bar chart). Enables like-for-like comparison across projects.
08 — Reporting Engine	Section I + all sections	Generates co-branded PDF disclosure reports aligned to a selected framework (GRI, ISSB, IFC, TNFD, or NUPRC). Nine-section report including indicators, regulations, stakeholders, mitigation measures, evidence, outcomes, impacts, and disclosure topics.

05 Reporting Engine & PDF Export

The Reporting Engine is the primary client-facing output of the platform. A user selects a project and a disclosure framework; the engine pulls data from all relevant sections and renders a live in-browser preview. The user can then export a print-quality PDF generated server-side using pdfkit — no Chromium, no headless browser dependency.

PDF report structure (9 sections)

§	SECTION	CONTENT
01	Project Overview	Operating environment, duration, prepared by
02	ESG Indicators Tracked	All Section B indicators for this project
03	Regulatory Compliance	All Section C regulations applied
04	Stakeholder Engagement	All Section D stakeholder groups and outcomes
05	Mitigation Measures & Decision Support	Section E — all measures with effectiveness and evidence
06	Regulatory Evidence	Section F — evidence submitted per regulation, acceptance status
07	Environmental Outcomes	Section G — outcomes, timeframe, measurement method, outstanding issues
08	Social Impacts	Section H — positive/negative impacts, grievance mechanism and outcome
09	Disclosure Topics	Section I filtered by selected framework (GRI/ISSB/IFC/TNFD/NUPRC)

Supported disclosure frameworks

CODE	FULL NAME	RELEVANCE TO NIGERIAN O&G SECTOR
GRI	Global Reporting Initiative	Universal sustainability disclosure standards (GRI 200/300/400). Most widely adopted globally.
ISSB	IFRS Sustainability Standards	IFRS S1 (general) and S2 (climate). Required for companies with listed parent companies or international investors.
IFC	IFC Performance Standards	Eight performance standards covering E&S risk, labour, community, land, biodiversity. Equator Principles banks require these.
TNFD	Nature-related Financial Disclosures	Nature, biodiversity, and ecosystem services risk disclosure. Emerging requirement for extractive sector.
NUPRC	NUPRC Environmental Guidelines	Nigerian Upstream Petroleum Regulatory Commission statutory annual environmental performance reporting.

06 Project Register & Extraction Workspace

Project Register

The project register is the platform's home screen. It displays all projects in a filterable table with search, tier, status, project type, and date filters. Each row shows the project name, reference code, type, location, data readiness tier, pipeline status, and document availability indicators (reports, monitoring data, GIS, photographs, drawings). A platform overview section above the table shows live statistics pulled from MongoDB.

Data Readiness Tiers

TIER	DEFINITION	PLATFORM TREATMENT
Tier 1 — Complete	Full documentation available and digitisation-ready	Priority for extraction; engines incorporate immediately
Tier 2 — Needs cleaning	Documentation exists but requires QA or gap-filling	Extractable with analyst review; flagged in register
Tier 3 — Reconstruction	Physical records only; scanning/transcription required	Held in inventory pending digitisation mobilisation

Extraction Workspace

Each project has a dedicated extraction workspace — a tabbed interface where analysts and reviewers fill in Sections A through L. The workspace shows a live completion progress bar, section-by-section status dots, and role-based access locks (a padlock icon on sections the current user cannot edit). All fields auto-save on blur. Sections with structured data (B, C, D, E, F, I) use repeatable row tables that grow as entries are added.

Guided data entry — helpers and quick-add panels

Each section includes contextual guidance to support analysts who may be unfamiliar with certain fields:

- Section B — collapsible "Quick-add common indicator" panel with 15 pre-defined E/S/G indicators for Nigerian oil & gas (TPH, PAH, DO, employment, fishing access, grievance rate, etc.). Click to insert a pre-filled row.
- Section C — "Quick-add common regulation" panel with 8 common Nigerian and international standards (NUPRC, EGASPIN, EIA Act, NOSDRA, IFC PS1/PS5/PS3, TNFD). Issuing body field has autocomplete with 13 common regulators.
- Section D — Stakeholder Group and Reporting Format fields both have autocomplete lists with 14 and 8 common options respectively.
- Section F — Issuing Body, Evidence Type, and Format/Frequency fields all have autocomplete suggestions.
- Section I — Framework reference cards explaining each framework's scope, plus a "Quick-add common topics" panel that shows standard disclosure topics per framework (GRI, ISSB, IFC, TNFD, NUPRC).

07

Administration & Platform Operations

Admin panel (admin role only)

The admin panel provides full user lifecycle management and data export capabilities:

- User table with inline name and role editing
- Inline password reset (admin sets a new password for any user)
- Create new user modal with email, name, role, and password fields
- Self-removal guard — admins cannot delete their own account
- Role summary cards showing counts per role
- CSV export: full project register (all fields) and indicator library (all Section B entries across all projects)
- System statistics dashboard — total projects, status breakdown, tier breakdown, extraction counts

Pipeline status workflow

STATUS	MEANING	WHO SETS IT
Inventoried	Project logged but not yet prioritised	Uptonville Reviewer on creation
Prioritised	Selected for extraction in this phase	Uptonville Reviewer
Digitising	Physical records being scanned/ prepared	Uptonville Reviewer
Extraction in progress	Template sections being filled	Set automatically when first section saved
Engine-mapped	Section K completed — all engines mapped	Analyst
Signed-off	All three parties signed Section L	Steering Committee

Data integrity utilities

- Backfill endpoint (POST /api/admin/backfill-section-a) — retroactively seeds Section A from the Project document for any existing projects where Section A was left empty at creation.
- Near-duplicate indicator detection — the Indicator Library engine uses Dice coefficient string similarity (>85% match threshold) to flag potentially duplicate indicator entries across projects.

08 Correspondence with the MoU Deliverables

The following table maps each key deliverable from the Strategic Collaboration Agreement and IIF Template document to the built component that delivers it.

MOU REQUIREMENT	BUILT COMPONENT	STATUS
Digitised project template with 12 structured sections (A–L)	ExtractionWorkspace with 12 tabbed sections, each with validated form fields	' Complete
Section A: Project Identification & Context	SectionA.tsx — name, ref code, type, location, env, client, operator, duration, value, tier	' Complete
Section B: ESG Indicator Library	SectionB.tsx — repeatable rows: indicator, E/S/ G, unit, method, why it mattered	' Complete
Section C: Regulatory Rules Engine input	SectionC.tsx — repeatable rows: category, regulation, issuing body, how applied	' Complete
Section D: Stakeholder Intelligence input	SectionD.tsx — repeatable rows: group, interest, reporting format, outcome	' Complete
Section E: Decision Support Engine input	SectionE.tsx — repeatable rows: measure, effectiveness, evidence, recommended?, reasoning	' Complete
Section F: Regulatory Evidence record	SectionF.tsx — repeatable rows: regulation, issuing body, evidence type, format, accepted?	' Complete
Section G: Environmental Outcomes	SectionG.tsx — outcomes achieved, timeframe, measurement method, outstanding issues	' Complete
Section H: Social Impacts	SectionH.tsx — positive/negative impacts, grievance mechanism and outcome	' Complete
Section I: Disclosure Alignment	SectionI.tsx — repeatable rows: topic, framework (GRI/ISSB/IFC/TNFD/NUPRC), why valuable	' Complete
Section J: Automation Opportunities backlog	SectionJ.tsx — data pain points, manual processes, automation opportunity, priority	' Complete
Section K: Engine Contribution Map	SectionK.tsx + EngineContributionMap model — tick per engine + most valuable insight	' Complete
Section L: Validation & Sign-off	SectionL.tsx + SignOff model — 3-party signature (Uptonville, Analyst, Steering Committee)	' Complete
Eight analytical engines aggregating across all projects	Eight aggregation pipelines in engines.controller.ts, eight engine screens in / engines	' Complete
Regulatory Rules Engine (C, F)	regulatoryRules() — cross-project regulation library with application context	' Complete
Indicator Library Engine (B)	indicatorLibrary() — with near-duplicate detection and filter by ESG/type	' Complete
Materiality Engine (B, D, G, H)	materiality() — stakeholder × project type × ESG category matrix	' Complete
Stakeholder Intelligence Engine (D)	stakeholderIntelligence() — recurring stakeholder profiles with format preferences	' Complete
Decision Support Engine (E)	decisionSupport() — searchable, filterable mitigation library with expert reasoning	' Complete
Benchmarking Engine (A, B)	benchmarking() — ESG coverage depth chart + unique indicators per project type	' Complete
Reporting Engine (I + all)	reportPreview() + generatePdf() — framework-filtered 9-section PDF report	' Complete

Co-branded outputs (Impact Driver × Uptonville)	All PDF headers, cover pages, and report footers carry both brand names	' Complete
GRI, ISSB, IFC, TNFD, NUPRC framework support	Framework selector in Reporting Engine; Section I aligned to all five	' Complete
Role-based access: Uptonville reviewer, Analyst, Steering Committee, Admin	Four roles with JWT auth, section-level write gates, middleware enforcement	' Complete
Data Readiness Tier 1/2/3 classification	Tier field on Project model; TierBadge UI component; filter in project register	' Complete
Admin user management	Admin panel: create, edit, reset password, delete users; CSV export	' Complete
Deployment-ready for production	render.yaml (backend) + vercel.json (frontend) + MongoDB Atlas	' Complete

09 Phase 2 — Recommended Next Scope

Phase 1 delivers the complete data ingestion, engine aggregation, and reporting infrastructure. The following capabilities are natural extensions for Phase 2, based on what emerged during the build and from patterns visible in the pilot data:

Data and content

- Ingest the remaining pilot projects (target: 7–10 total). The platform is live and ready — this is a data entry exercise, not a development one. With 3+ projects in the engines, the benchmarking and materiality matrices become meaningful.
- GIS integration — link Section A location data to a map view showing project sites across Nigeria, filterable by type and tier.
- Document attachment — allow PDF reports and monitoring data files to be attached directly to project records rather than referenced externally.

Platform capability

- Automated regulatory report generation — use Section C/F data to auto-populate NUPRC e-portal submission templates.
- IoT sensor integration pathway — Section J consistently identifies real-time monitoring as the highest-priority automation. A data ingestion API for continuous monitoring data (TPH proxy, DO, turbidity) would close this gap.
- Multi-language support — community-facing reporting in local languages (Ibibio, Ijaw, Yoruba, Hausa) was flagged in Section D stakeholder entries as a recurring need.
- Carbon accounting module — scope 1/2/3 emissions tracking linked to Section B GHG indicators, feeding ISSB S2 disclosure automatically.

Commercial

- White-label packaging for other Nigerian IOC contractors — the platform architecture is generic enough to onboard additional operators with their own project portfolios.
- API access tier — allow client organisations to query the Indicator Library and Regulatory Rules Engine programmatically, enabling integration with their own project management systems.

BUILT ON

Impact Intelligence Framework

A co-branded platform by Impact Driver and Uptonville Nigeria Limited
for ESG intelligence in the Nigerian upstream petroleum sector.

Phase 1 build completed 02 July 2026.